

MINI PROJECT-2

MOBILE PRICE PREDICTION

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INTRODUCTION

- In today's fast-paced world, mobiles have become an integral part of our lives. With numerous options available, consumers often face the challenge of selecting the right mobile device within their budget.
- It is also critical for the mobile manufacturers to put a reasonable and affordable price for the respective features as it directly impacts the profit of the manufacturer.
- Machine Learning algorithms offer a solution by predicting mobile prices based on various features and specifications.
- We use some popular algorithms like KNN, Logistic Regression, Decision Tree, and Random Forest and adapt the suitable one to predict mobile prices accurately.



EXISTING SYSTEM

- The existing system to predict the mobile prices is companies relying on human experts who analyze the specifications and features of the mobile and analysis of various statistical models is done.
- Time-consuming
- Prone to errors
- Not reliable
- Not efficient
- Prediction may be subjective



PROBLEM STATEMENT



To create a Machine Learning model that is efficient and fast in predicting the price of the mobile accurately and building an interface to show the outputs to the viewers



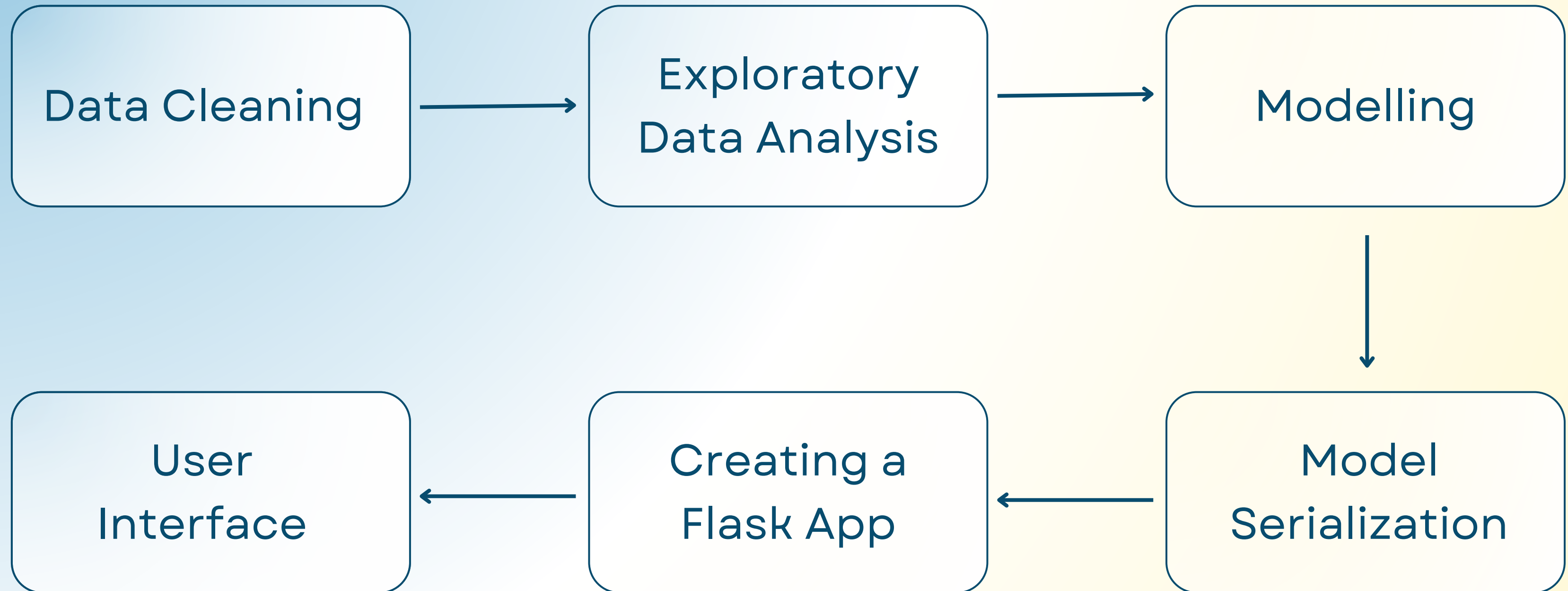
PROPOSED SYSTEM



- The proposed system is providing an interface for the sellers to input the specifications and features of their mobile and get accurate price predictions.
- Increase the efficiency of the predicting process
- Reduce the time consumed to predict the prices
- Make the system reliable for sellers
- Improve the seller's competitiveness
- Increase the seller's profitability



METHODOLOGY





SOFTWARE REQUIREMENTS

- **Programming languages:** Python
- **Data Analysis and Visualization libraries:** NumPy, Pandas, Matplotlib
- **Modelling libraries:** scikit-learn, pickle
- **HTTP Server:** Gunicorn

Front-End:

- HTML
- CSS
- Bootstrap

Back-End:

- Flask



DATASET USED



The mobile price dataset was downloaded from kaggle

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	battery_pcm	blue	clock_speed	dual_sim	fc	four_g	int_memo	m_dep	mobile_wt	n_cores	pc	px_height	px_width	ram	sc_h	sc_w	talk_time	three_g	touch_screen	wifi	price_range
2	842	0	2.2	0	1	0	7	0.6	188	2	2	20	756	2549	9	7	19	0	0	1	1
3	1021	1	0.5	1	0	1	53	0.7	136	3	6	905	1988	2631	17	3	7	1	1	0	2
4	563	1	0.5	1	2	1	41	0.9	145	5	6	1263	1716	2603	11	2	9	1	1	0	2
5	615	1	2.5	0	0	0	10	0.8	131	6	9	1216	1786	2769	16	8	11	1	0	0	2
6	1821	1	1.2	0	13	1	44	0.6	141	2	14	1208	1212	1411	8	2	15	1	1	0	1
7	1859	0	0.5	1	3	0	22	0.7	164	1	7	1004	1654	1067	17	1	10	1	0	0	1
8	1821	0	1.7	0	4	1	10	0.8	139	8	10	381	1018	3220	13	8	18	1	0	1	3
9	1954	0	0.5	1	0	0	24	0.8	187	4	0	512	1149	700	16	3	5	1	1	1	0
10	1445	1	0.5	0	0	0	53	0.7	174	7	14	386	836	1099	17	1	20	1	0	0	0
11	509	1	0.6	1	2	1	9	0.1	93	5	15	1137	1224	513	19	10	12	1	0	0	0
12	769	1	2.9	1	0	0	9	0.1	182	5	1	248	874	3946	5	2	7	0	0	0	3
13	1520	1	2.2	0	5	1	33	0.5	177	8	18	151	1005	3826	14	9	13	1	1	1	3
14	1815	0	2.8	0	2	0	33	0.6	159	4	17	607	748	1482	18	0	2	1	0	0	1
15	803	1	2.1	0	7	0	17	1	198	4	11	344	1440	2680	7	1	4	1	0	1	2
16	1866	0	0.5	0	13	1	52	0.7	185	1	17	356	563	373	14	9	3	1	0	1	0
17	775	0	1	0	3	0	46	0.7	159	2	16	862	1864	568	17	15	11	1	1	1	0
18	838	0	0.5	0	1	1	13	0.1	196	8	4	984	1850	3554	10	9	19	1	0	1	3
19	595	0	0.9	1	7	1	23	0.1	121	3	17	441	810	3752	10	2	18	1	1	0	3
20	1131	1	0.5	1	11	0	49	0.6	101	5	18	658	878	1835	19	13	16	1	1	0	1
21	682	1	0.5	0	4	0	19	1	121	4	11	902	1064	2337	11	1	18	0	1	1	1
22	772	0	1.1	1	12	0	39	0.8	81	7	14	1314	1854	2819	17	15	3	1	1	0	3
23	1709	1	2.1	0	1	0	13	1	156	2	2	974	1385	3283	17	1	15	1	0	0	3
24	1949	0	2.6	1	4	0	47	0.3	199	4	7	407	822	1433	11	5	20	0	0	1	1
25	1602	1	2.8	1	4	1	38	0.7	114	3	20	466	788	1037	8	7	20	1	0	0	0



ALGORITHMS USED



1. KNN
2. Logistic Regression
3. Decision Tree
4. Random Forest



RESULTS



KNN

It provided a score of

92.1

LOGISTIC REGRESSION

It provided a score of

74.6

DECISION TREE

It provided a score of

81.6

RANDOM FOREST

It provided a score of

87.5



OUTPUT



127.0.0.1:5000

Mobile Price Predictor

Enter the battery power in mAh

Does it support bluetooth?

☐ Yes

☐ No

Enter the clock speed(the speed at which microprocessor executes instructions)

Does it support dual sim?

☐ Yes

☐ No

Enter the front camera mega pixels:

Enter the primary camera mega pixels:

Enter the pixel resolution height:

Enter the pixel resolution width:

Does it support 5G?

☐ Yes

☐ No

Enter the internal memory in gigabytes:

Enter the mobile depth in cm:

Enter the mobile weight in gms:

Enter the number of cores in the mobile:

Enter the Screen Width of mobile in cm

Enter the longest time that a single battery charge will last:

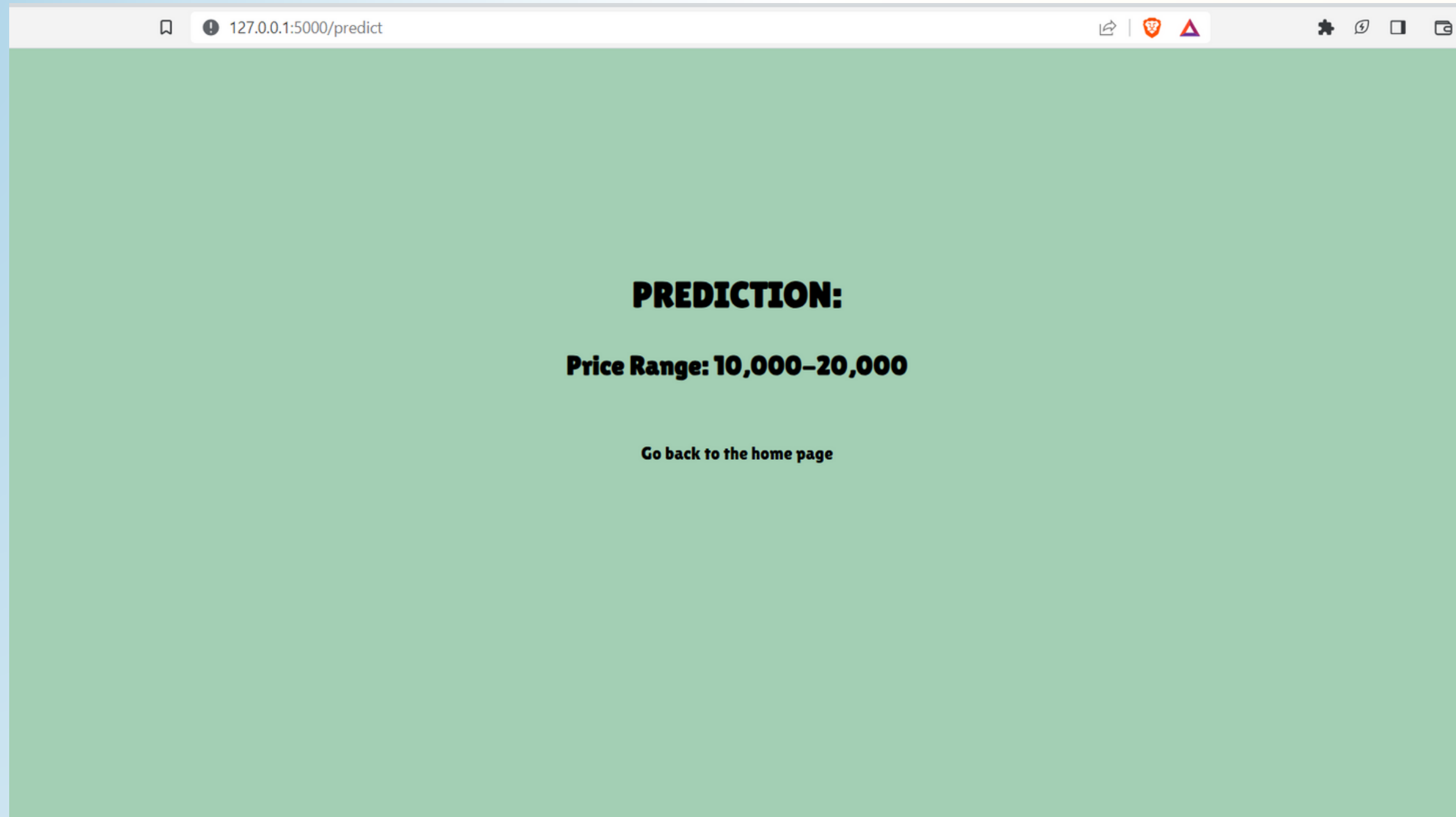
Does it support 4G?

☐ Yes

☐ No

Windows taskbar: Search, Edge, File Explorer, Microsoft Store, Visual Studio Code, PowerPoint, Chrome, Docker Desktop, Telegram, Discord, Spotify, RStudio, System tray: Network, Volume, ENG IN, 14

OUTPUT





CONCLUSION

In conclusion, our project successfully combined the power of machine learning models with the flexibility of the Flask web framework to create an interactive and user-friendly webpage. We achieved the following key milestones:

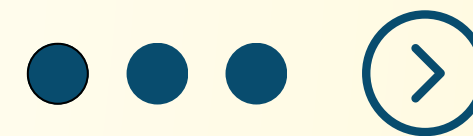
- Model Development and Training
- Serialization and Model Persistence
- Flask Web Application
- Accurately predicting and displaying the price ranges



FUTURE SCOPE

Looking ahead, there are several exciting possibilities for further improving our project:

- Enhanced User Experience
- Model Refinement and Interpretability
- Reducing the input intake
- Integration with Real-Time Data



THANK
YOU